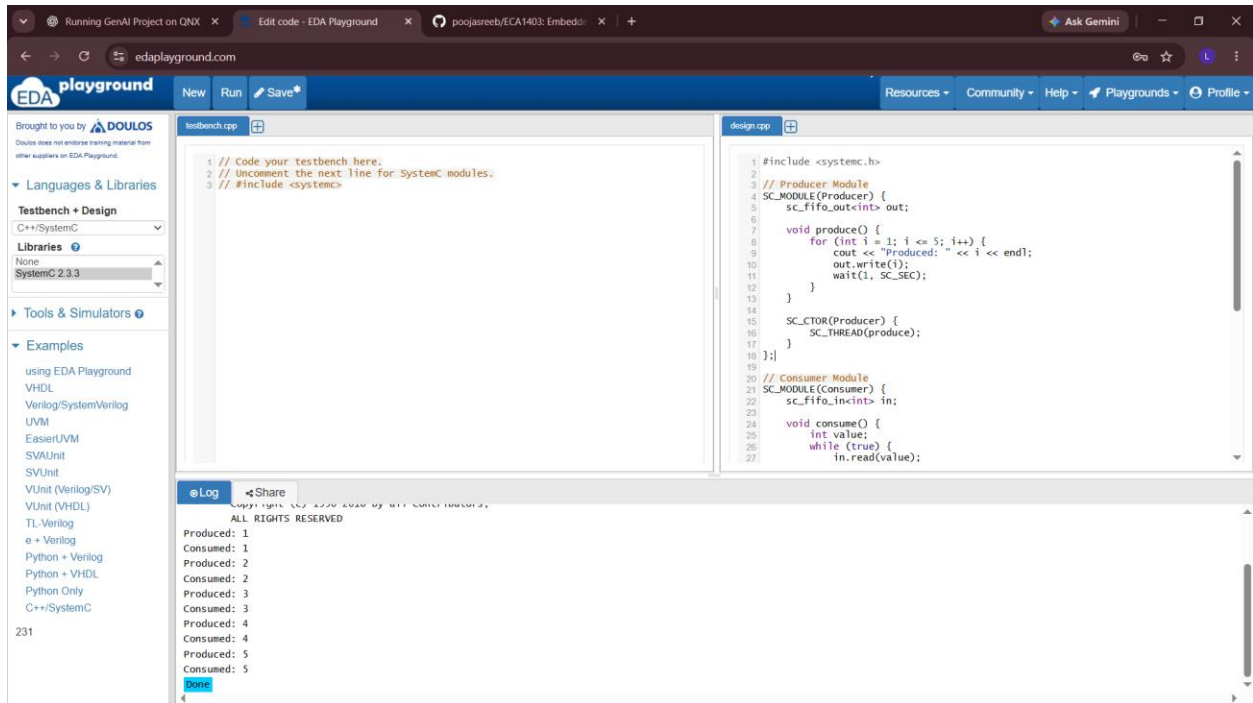




SOURCE CODE

```
#include <systemc.h>
```

```
// Producer Module
```

```
SC_MODULE(Producer) {
    sc_fifo_out<int> out;
```

```
    void produce() {
        for (int i = 1; i <= 5; i++) {
            cout << "Produced: " << i << endl;
            out.write(i);
            wait(1, SC_SEC);
        }
    }
};
```

```
    SC_CTOR(Producer) {
        SC_THREAD(produce);
    }
};
```

```
// Consumer Module
```

```
SC_MODULE(Consumer) {
    sc_fifo_in<int> in;
```

```
    void consume() {
        int value;
```

```
        while (true) {
            in.read(value);
            cout << "Consumed: " << value << endl;
            wait(1, SC_SEC);
        }
    }

    SC_CTOR(Consumer) {
        SC_THREAD(consume);
    }
};

int sc_main(int argc, char* argv[]) {
    sc_fifo<int> fifo(5);

    Producer producer("Producer");
    Consumer consumer("Consumer");

    producer.out(fifo);
    consumer.in(fifo);

    sc_start(10, SC_SEC);

    return 0;
}
```